

Academic CV

RESEARCH PROFILE

MPhil researcher at the University of Sydney building reliable, efficient AI systems across federated learning, semantic communication, and trustworthy machine learning.

RESEARCH INTERESTS

Reliable machine learning · Federated learning · Semantic communication · Efficient AI systems · Distributed optimisation · Generative AI systems

APPOINTMENTS AND EDUCATION

Master of Philosophy in Computer Science

[The University of Sydney](#) | 2025 - present | Sydney, Australia

Research in reliable and efficient AI systems, supervised within the DUAL Group.

- Research across federated learning, semantic communication, calibration, and resource-constrained AI.
- Build reproducible methods and experimental pipelines from literature review through analysis.

Casual Academic

[UNSW Sydney](#) | Aug 2026 - present | Sydney, Australia

Teaching and academic support in computing coursework.

- Facilitate classes and support students through technical concepts and practical problem solving.
- Apply rubric-aligned feedback and clear explanations to student work.

Casual Academic Tutor

[The University of Sydney](#) | Feb 2026 - present | Sydney, Australia

Tutorial delivery and student learning support for university coursework.

- Deliver weekly tutorials, class activities, small-group discussions, and question sessions.
- Support academic communication, teamwork, technical problem solving, and professional practice.
- Coordinate with unit teams to deliver consistent, inclusive learning experiences.

Research Assistant

[DUAL Group, The University of Sydney](#) | Apr 2024 - present | Sydney, Australia

Research engineering and experimentation for distributed learning and efficient AI systems.

- Build reproducible training and evaluation pipelines with explicit configs, seeds, logging, and reporting.
- Implement baselines, targeted ablations, and publication-quality figures and tables.
- Coordinate experiment ownership and artifact handoff across code, plots, and manuscripts.
- Designed and deployed the official DUAL Group website.

Bachelor of Advanced Computing, Data Science

[The University of Sydney](#) | 2021 - 2025 | Sydney, Australia

Graduated with Distinction; final-year work centred on machine learning and distributed systems.

- Undergraduate thesis in adaptive low-rank optimisation for federated learning, marked 85/100.
- Coursework included machine learning, deep learning, natural language processing, and distributed systems.

PUBLICATIONS

BRAVE: Block-wise Structural Regularization via Controlled Evidence Feedback for Reliable Label Aggregation under Sparse Crowdsourcing

Transactions on Machine Learning Research, 2026 | under review

Zerun Niu, et al.

Distributionally Robust Wireless Semantic Communication with Large AI Models

IEEE Journal on Selected Areas in Communications, 2026 | published

L. T. Le, S. H. Wanasekara, Zerun Niu, et al.

Federated Deep Equilibrium Learning over Resource-Constrained Edge Networks

IEEE Internet of Things Journal submission, 2025 | submitted

L. T. Le, Zerun Niu, T. D. Nguyen, et al.

SELECTED RESEARCH PROJECTS

Distributionally Robust Wireless Semantic Communication with Large AI Models

2025-2026 | published

Contributed modelling, experiment design, robust training simulations, and evaluation under constrained channels. Evidence: JSAC 2026 publication venue.

Federated Deep Equilibrium Learning over Resource-Constrained Edge Networks

2025-present | submitted

Implemented FeDEQ components, explored communication-efficient aggregation, and evaluated non-IID NLP and vision settings. Evidence: NLP + vision evaluation domains.

BRAVE: Block-wise Structural Regularization via Controlled Evidence Feedback for Reliable Label Aggregation under Sparse Crowdsourcing

2026 | under review

Zerun Niu - first author; led algorithm design, literature review, experimental design, code implementation, and experimental deployment. Evidence: 14 crowdsourcing benchmarks; 5/14 lowest NLL; 9/14 best or tied-best ECE; 11/14 within 0.03 accuracy of the strongest baseline.

DUAL Group Research Website

2024-present | active

Designed, developed, deployed, and maintain the group's official web presence and research communication system. Evidence: Design -> deploy end-to-end ownership.

Adaptive Low-Rank Optimisation for Federated Learning

2025 | completed

Undergraduate thesis; designed and evaluated adaptive low-rank optimisation methods. Evidence: 85/100 thesis mark.

AI Study Assistant

2024 | completed

Designed and implemented the end-to-end application and retrieval workflow.

Vision Transformer Joint Source-Channel Coding

2024 | completed

Implemented and evaluated model variants across changing channel conditions.

Academic CV

TEACHING

Casual Academic

UNSW Sydney | Aug 2026 - present

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Casual Academic Tutor

The University of Sydney | Feb 2026 - present

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TECHNICAL PRACTICE

Research methods

Methods and tools

Algorithm design · Literature synthesis · Experimental design · Ablation studies · Calibration · Robustness evaluation

ML engineering

Methods and tools

Python · PyTorch · Hugging Face · scikit-learn · Reproducible pipelines · Experiment tracking

Distributed systems

Methods and tools

Federated learning · Ray · Dask · Spark · Airflow · FastAPI

GenAI

Methods and tools

LLM and VLM orchestration · RAG · Agents · LoRA and PEFT · Evaluation · Safety guardrails

Web and research tooling

Methods and tools

TypeScript · React · Astro · GitHub Actions · Data visualisation · Research communication

PUBLIC LINKS

Website: zerunniu.github.io

OpenReview (BRAVE): openreview.net/forum?id=iWFI5hO1dZ